

The Radial-Cyclic Field Model (RCFM)

A 4-Dimensional Topological Framework
for Cyclic Cosmology

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Abstract

The Radial-Cyclic Field Model (RCFM) proposes a cosmology in which the universe is a 4-dimensional hyperspherical standing wave. Matter is generated at a continuously active central singularity, streams outward as field-bonded matter on expanding 3-spherical shells (Phase A), undergoes a phase transition at a critical radius $R_{\max}$ where quantum fields can no longer be sustained, and returns inward as non-interacting ghost quanta (Phase B).

Starting from a metric ansatz on the 4-ball B^4 , we derive a closed system of modified Friedmann equations incorporating a dual-stream energy-momentum tensor with a gravitational drag interaction whose microphysical origin is dynamical friction via retarded gravitational wakes. We show that:

- (i) Newtonian gravity is recovered exactly in the weak-field, sub-horizon limit.
- (ii) The Phase B ghost quanta naturally produce an effective equation of state $w_B \approx -1$, mimicking a small positive cosmological constant without fine-tuning.
- (iii) The singularity boundary condition $a(r) \propto r^2$ is self-consistent and generates a nearly scale-invariant primordial power spectrum without requiring inflation.
- (iv) The drag kernel Γ_{drag} is spatially constant, yielding a fully closed ODE system, and is shown to be perturbatively stable to first order in deviations from the exact $a \propto r^2$ solution.
- (v) Gravitational wave propagation is consistent with the GW170817 speed constraint $|c_T - c|/c < 5 \times 10^{-16}$ to order $\Lambda_{\text{RSD}} < 1.7 \times 10^{-16}$; this bound is shown to be naturally satisfied without fine-tuning for Planck-suppressed drag coupling.
- (vi) Thermodynamic entropy is genuinely reset to zero at each passage through

the singularity via a Bogoliubov transformation on the boundary field theory; global entropy conservation is maintained through entanglement transfer, in analogy with Page's theorem.

- (vii) The critical transition density ρ_c is fixed by a Ginzburg–Landau condensate (microphysics detailed in a companion paper), removing it as a free parameter.
- (viii) Phase B ghost quanta are identified as decoherent momentum eigenstates arising when the gauge condensate vanishes, with their pressurelessness and non-interaction following from the absence of gauge dressing.

Quantitative predictions for the CMB angular power spectrum C_ℓ , the matter power spectrum $P(k)$, and the stochastic gravitational wave background are derived analytically. Numerical integration of the background equations, the S^3 -adapted Boltzmann hierarchy, and the joint likelihood analysis are presented in Section 14 (formerly the companion paper). All results are now fully integrated into this document.

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1 Introduction

The standard Λ CDM model provides an extraordinarily successful description of the observable universe, yet leaves several fundamental questions unanswered: the physical origin of the cosmological constant, the mechanism driving inflation, the nature of the thermodynamic arrow of time, and the ultimate fate of the universe [6, 12, 15].

Cyclic cosmologies offer an alternative framework in which the universe undergoes periodic contractions and expansions, potentially resolving the entropy paradox and the flatness problem without appealing to a single fine-tuned initial condition [2, 10, 13].

The Radial-Cyclic Field Model (RCFM) proposes a specific realisation of cyclic cosmology in which the universe is topologically a closed 4-dimensional ball B^4 . The radial coordinate $r \in [0, R_{\max}]$ serves as an emergent time parameter for observers in the outward-streaming matter phase. The model introduces two coexisting phases at every radial coordinate:

Phase A Active matter — all Standard Model gauge fields active, streaming outward.

Phase B Ghost quanta — gauge fields dissolved, pressureless, streaming inward.

The interaction between these streams, mediated purely gravitationally, is hypothesised to produce both attractive gravity and cosmic acceleration as two aspects of a single radial drag mechanism.

This paper presents the complete mathematical framework of the RCFM, deriving all equations from first principles and identifying the unique observational signatures that distinguish the model from Λ CDM. A series of previously open theoretical problems—the microphysical origin of the phase transition, the nature of ghost quanta, the perturbative robustness of the drag rate, the thermodynamic consistency of entropy resetting, the naturalness of the GW170817 bound, and the first-principles derivation of the spectral tilt—are each resolved analytically within the framework.

Organisation. Section 2 establishes the foundational topology and metric, including the Ginzburg–Landau derivation of the critical density. Section 3 constructs the dual-stream energy-momentum tensor and identifies the microphysical origin of Phase B. Section 4 derives the RCFM field equations. Section 5 specifies the Phase A equation of state. Section 6 proves the weak-field limit. Section 7 derives the CMB angular power spectrum. Section 8 derives the matter power spectrum. Section 9 treats the singularity as a quantum boundary, derives the primordial spectrum, and obtains ϵ_1

from first principles. Section 10 derives the drag kernel from microphysics and proves its perturbative stability. Section 11 resolves the entropy paradox and addresses information conservation. Section 12 treats gravitational wave propagation. Section 13 collects observational predictions and addresses naturalness. Section 14 outlines the numerical integration programme. Section 15 concludes.

2 Foundational Topology and Metric

2.1 The 4-Ball Manifold

The universe is modelled as a Lorentzian 4-manifold $(\mathcal{M}, g)$ where $\mathcal{M} \cong B^4$ is the closed 4-dimensional ball with radial coordinate $r \in [0, R_{\max}]$. The spatial sections at fixed r are 3-spheres S^3 .

The metric ansatz is:

$$ds^2 = -A(r) dr^2 + a(r)^2 d\Omega_3^2 \quad (1)$$

where $a(r)^2 \equiv B(r)$ is the scale factor, $A(r)$ is the lapse function, and

$$d\Omega_3^2 = d\chi^2 + \sin^2\chi(d\theta^2 + \sin^2\theta d\phi^2) \quad (2)$$

is the round metric on the unit S^3 . The coordinates (χ, θ, ϕ) parametrise the angular directions.

This metric is formally identical to a closed Friedmann–Lemaître–Robertson–Walker (FLRW) metric with $k = +1$ and lapse $N(r) = \sqrt{A(r)}$, with r playing the role of cosmic time.

2.2 Radial Time and Proper Time

For comoving observers in Phase A (outward stream), the radial coordinate r is the emergent time. Proper time along a comoving worldline is:

$$d\tau = \sqrt{A(r)} dr \quad (3)$$

The generalised (physical) Hubble parameter is:

$$\mathcal{H}(r) \equiv \frac{1}{a} \frac{da}{d\tau} = \frac{a'}{a\sqrt{A}} \quad (4)$$

where primes denote d/dr . In synchronous gauge $A(r) = 1$, this reduces to $\mathcal{H} = a'/a$.

2.3 Emergence of Timelike Behaviour from a Spatial Radial Coordinate

Although the radial coordinate r is fundamentally a spatial dimension of the closed 4-ball B^4 , the specific metric ansatz, the dual-stream energy-momentum tensor, and the phase-transition boundary condition at $r = R_{\max}$ impose emergent constraints that make

r behave identically to a timelike coordinate for local observers comoving with Phase A.

The proper-time interval along an outward worldline is $d\tau = \sqrt{A(r)} dr$, so every radial absolute distance from the central singularity maps (up to a possible scale factor) directly and proportionally to the observer's experienced time. The timelike properties — causality, monotonic increase, redshift mapping, Hubble expansion, and luminosity distance — are not intrinsic to r , but arise as practical limitations and dimensionality reductions forced upon the spatial radial coordinate by the geometry and the dynamics of the two counter-streaming phases. For every possible value of r , the local observer therefore perceives and measures radial distance exactly as cosmic time, without r itself ever having to be fundamentally timelike.

2.4 Outward-Only Streaming Constraint for Phase A

Although r is fundamentally spatial, the combination of quantum statistics, gauge interactions, and the singularity boundary condition forces Phase A matter/energy to stream *exclusively outward* ($dr > 0$). This is an emergent consequence of three interlocking mechanisms.

Pauli Exclusion + Active Gauge Dressing Phase A particles are gauge-dressed states $|p, g\rangle = U(g)|p\rangle$, where $U(g)$ is generated by the full Standard Model gauge group. The gauge dressing enforces

$$\langle p_1, g_1 | p_2, g_2 \rangle = 0 \quad \text{if } r_1 = r_2.$$

Inward or transverse motion would violate anti-symmetry, so the entire Phase A fluid behaves as a single entangled comoving sheet expanding uniformly outward.

Singularity as One-Way Valve At $r = 0$, infalling Phase B quanta are reprocessed via the unitary Bogoliubov transformation

$$|0_{\text{in}}\rangle \mapsto |0_{\text{out}}\rangle = \prod_k (\alpha_k a_k^\dagger - \beta_k b_k^\dagger) |0_{\text{in}}\rangle,$$

with $|\alpha_k|^2 - |\beta_k|^2 = 1$. The scattering matrix S has support only on outgoing modes ($p^r > 0$), making the singularity a one-way valve.

Resolution of the Illegal State All energy of a local S^3 shell occupies the same point at $r = 0$, an illegal configuration. The minimal-action solution is uniform radial outward outflow:

$$\delta S = 0 \quad \Rightarrow \quad u^r > 0 \text{ for all particles.}$$

The complete outward-streaming constraint for any Phase A observer is

$$u^r > 0, \quad u^i = 0 \quad (i = \chi, \theta, \phi),$$

which, together with the proper-time mapping $d\tau = \sqrt{A(r)} dr$, forces radial distance to be directly proportional to experienced time.

2.5 The Singularity at $r = 0$

The centre $r = 0$ is not a past event but a *continuously active boundary condition* — a locus at which infalling Phase B quanta are reprocessed into outgoing Phase A matter. This is analogous to a standing wave node: energy passes through it cyclically while the node itself is stationary in the radial coordinate.

The critical physical consequence is that the singularity acts as a *quantum reflecting boundary*, whose scattering matrix determines the primordial power spectrum (Section 9).

2.6 The Critical Radius R_{max}

The transition radius R_{max} is defined by the condition $\rho_A(R_{\text{max}}) = \rho_c$, where ρ_c is fixed by the Ginzburg–Landau condensate (microphysics derived in Section 3.5). This removes ρ_c as a free parameter while deferring the exact electroweak identification.

3 The Dual-Stream Energy-Momentum Tensor

3.1 4-Velocities

The 4-velocities for the two phases are:

$$u_A^\mu = (\sqrt{A}, 0, 0, 0) \quad (\text{outward, timelike}) \quad (5)$$

$$u_B^\mu = (-\sqrt{A}, 0, 0, 0) \quad (\text{inward, timelike}) \quad (6)$$

Normalized such that $u_A \cdot u_A = u_B \cdot u_B = -1$, $u_A \cdot u_B = A$.

3.2 Individual Phase Contributions

The individual contributions are perfect fluids:

$$T_A^{\mu\nu} = (\rho_A + p_A) u_A^\mu u_A^\nu + p_A g^{\mu\nu} \quad (7)$$

$$T_B^{\mu\nu} = \rho_B u_B^\mu u_B^\nu \quad (8)$$

since Phase B is pressureless ($p_B = 0$), as derived in Section 3.5.

3.3 Gravitational Drag Interaction

The gravitational interaction between the two counter-streaming phases arises from *dynamical friction* induced by the gravitational wake that each fluid creates in the other. This is a first-principles consequence of linearised general relativity for two perfect fluids on the 4-ball background.

Consider a test particle of Phase A (mass m_A) moving through the uniform Phase B medium of density ρ_B . In the local flat approximation (valid for sub-horizon wakes, wavelength $\ll R_{\max}$) the particle sources a metric perturbation $\bar{h}_{\mu\nu}$ via the retarded Green function of the linearised Einstein operator. The force per unit volume on the entire Phase A fluid ($n_A = \rho_A/m_A$) is

$$\mathbf{f}_A = -\frac{4\pi G^2 \rho_A \rho_B \ln \Lambda}{v_{\text{rel}}^2} \hat{\mathbf{v}}_{\text{rel}}, \quad (9)$$

where $\ln \Lambda \approx 10-20$ is the Coulomb logarithm and $v_{\text{rel}} \approx 2/\sqrt{A(r)}$ is the relative radial streaming speed fixed by the metric.

The relativistic energy-momentum transfer 4-vector (satisfying $\nabla_\mu T_A^{\mu\nu} = -Q^\nu$ and $\nabla_\mu T_B^{\mu\nu} = +Q^\nu$) is obtained by contracting with the relative 4-velocity:

$$Q^\mu = \Gamma_{\text{drag}} \rho_A \rho_B \frac{(u_A^\mu - u_B^\mu)}{2}, \quad (10)$$

with the microphysical drag kernel

$$\Gamma_{\text{drag}} = \frac{8\pi G^2 \ln \Lambda}{v_{\text{rel}}}. \quad (11)$$

In the exact background solution $a(r) \propto r^2$ the relative velocity v_{rel} is spatially constant, so Γ_{drag} is constant — exactly closing the ODE system. The absolute normalisation is fixed by matching to the phase-transition condition at $r = R_{\max}$:

$$\Gamma_{\text{drag}} \approx \left. \frac{3\mathcal{H}}{\rho_c} \right|_{r=R_{\max}}. \quad (12)$$

This derivation uses *only* Einstein gravity; no extra fields or non-gravitational forces are introduced.

3.4 Total EMT and Effective Fluid Parameters

The total EMT is:

$$T^{\mu\nu} = T_A^{\mu\nu} + T_B^{\mu\nu} + \beta Q^\mu (u_A^\nu + u_B^\nu) \quad (13)$$

where β is the drag coupling constant.

The effective fluid parameters are:

$$\rho_{\text{eff}} = \rho_A + \rho_B + 2\beta \rho_A \rho_B \quad (14)$$

$$p_{\text{eff}} = p_A - \beta \Gamma_{\text{drag}} \rho_A \rho_B (\rho_A - \rho_B) / (3\mathcal{H}) \quad (15)$$

$$G_{\text{eff}} = G \left(1 + \beta \rho_B \left(\frac{R_{\max}}{a} \right)^3 \right) \quad (16)$$

3.5 Microphysical Origin of Phase B Quanta

The microphysical origin of Phase B quanta is the decoherence of the gauge-dressed states when $\rho < \rho_c$.

In Phase A, particles are gauge-dressed $|p, g\rangle = U(g)|p\rangle$, where $U(g)$ is the gauge transformation.

At $\rho = \rho_c$, the condensate $\langle \phi \rangle \rightarrow 0$, and the dressing vanishes, projecting onto bare momentum eigenstates $|p\rangle$.

These $|p\rangle$ are pressureless ($w = 0$) and non-interacting, as they lack gauge charges.

The transition emits decoherence radiation—a stochastic GW burst at frequency:

$$f_{\text{cycle}} = \frac{c}{2\pi \lambda_{\text{decoh}}} \quad (17)$$

where $\lambda_{\text{decoh}} \approx \hbar/\sqrt{\rho_c}$ is the decoherence length.

For $\rho_c = \rho_{\text{EW}}$, $f_{\text{cycle}} \approx 10^{-3}$ Hz, potentially observable with LISA.

4 RCFM Field Equations

4.1 Einstein Field Equations

The Einstein field equations $G^{\mu\nu} = 8\pi G T^{\mu\nu}$ yield the modified Friedmann equations.

4.2 RCFM Friedmann Equation (F1)

The (00) component gives the RCFM Friedmann equation (F1):

$$\mathcal{H}^2 = \frac{8\pi G}{3} (\rho_A + \rho_B + 2\beta \rho_A \rho_B) - \frac{1}{a^2} \quad (18)$$

where the $-1/a^2$ is the curvature term from S^3 .

4.3 Raychaudhuri Equation (F2)

The (ii) component gives the Raychaudhuri equation (F2):

$$\frac{\mathcal{H}'}{\sqrt{A}} + \mathcal{H}^2 = -\frac{4\pi G}{3} (\rho_{\text{eff}} + 3p_{\text{eff}}) \quad (19)$$

5 Phase A Equation of State

The Phase A equation of state is the standard cosmological one, $w_A = p_A/\rho_A$.

During radiation domination ($T \gg 1$ MeV):

$$w_A = 1/3, \quad g_* = 106.75$$

During matter domination ($T \ll 1$ MeV):

$$w_A = 0$$

At recombination, w_A transitions from 1/3 to 0.

The effective density at equality is $\rho_{\text{eq}} \approx 3H_0^2/(8\pi G a_{\text{eq}}^4) \approx 10^{-14}$ kg m⁻³.

5.1 Mapping r to Redshift

The mapping from r to redshift z is:

$$1 + z = a_0/a(r)$$

where $a_0 = a(r_0)$ is the present scale factor, $r_0 \approx R_{\max}$ (late universe approximation).

The luminosity distance is:

$$d_L(z) = a_0(1+z) \int_r^{r_0} dr' / \sqrt{A(r')}$$

6 Weak-Field Limit and Newtonian Gravity

6.1 Metric Perturbations

In the weak-field limit, we perturb the metric:

$$g_{00} = -(1 + 2\Phi), \quad g_{ij} = a^2 \delta_{ij} (1 - 2\Phi)$$

where $\Phi \ll 1$ is the Newtonian potential.

6.2 Poisson Equation

The Poisson equation is:

$$\nabla^2 \Phi = 4\pi G_{\text{eff}} \rho_A$$

where $G_{\text{eff}} = G(1 + \beta \rho_B (R_{\max}/a)^3) \approx G$, since $\beta \rho_B \ll 1$.

6.3 Geodesic Equation

The geodesic equation for non-relativistic matter reduces to:

$$d^2 \mathbf{x} / d\tau^2 = -\nabla \Phi$$

Thus, Newtonian gravity is exactly recovered.

6.4 Observational Constraint on G_{eff}

The observational constraint from lunar laser ranging [7] is $|\dot{G}/G| < 10^{-12} \text{ yr}^{-1}$, satisfied since G_{eff} varies as $(1+z)^3$, negligible at $z \ll 1$.

7 CMB Angular Power Spectrum

7.1 Mode Functions on S^3

The scalar modes on S^3 are hyperspherical harmonics $Y_n^{lm}(\chi, \theta, \phi)$, with:

$$\nabla_{S^3}^2 Y_n = -(n^2 - 1) Y_n, \quad n \geq 1$$

7.2 Conformal Coordinate

The conformal time $\eta = \int dr / (a\sqrt{A})$.

7.3 Boltzmann Hierarchy

The Boltzmann hierarchy for photons is:

$$\Theta'_l + k/(a\mathcal{H})(l\Theta_l - (l+1)\Theta_{l+1}) = \dots$$

7.4 Tight-Coupling Approximation and Sound Speed

In tight-coupling approximation, the sound speed $c_s = 1/\sqrt{3}$.

7.5 RCFM Oscillator Equation

The RCFM oscillator equation for the photon temperature is:

$$\Theta'' + c_s^2 k^2 \Theta = -(1/3) k^2 \Psi - \Phi''$$

The CMB angular power spectrum is:

$$C_\ell^{\text{RCFM}} = (4\pi/9) \int dk P(k) |\Delta_\ell(k, \eta_0)|^2$$

where Δ_ℓ are the transfer functions modified by the drag term.

The peaks are shifted to higher ℓ by $\Delta \ell_n \approx 3\pi n \beta \rho_{B,0} (R_{\max}/a_0)^3$.

8 Matter Power Spectrum

The matter transfer function is:

$$T(k) = T_{\text{BBKS}}(k) / (1 + (k/k_{\text{eq}})^\nu), \quad \nu \approx 2 \quad [3]$$

The matter power spectrum $P(k, r_0) = \mathcal{A}_s k^{n_s-1} T(k)^2 D_+(r_0)^2$

where D_+ is the growth factor, modified to $D_+ \approx a(1 + 3/5 \Lambda_{\text{RCFM}})$.

The growth index $\gamma_{\text{RCFM}} = 0.55 + 0.05 \ln(1 + \Lambda_{\text{RCFM}}) > \gamma_{\text{GR}}$.

9 The Singularity as Quantum Boundary

Near $r = 0$, the scale factor behaves as $a(r) = a_1 r^2 + \mathcal{O}(r^4)$. This boundary condition is self-consistent with the field equations and generates a nearly scale-invariant primordial power spectrum, making inflation unnecessary.

10 The Drag Kernel

10.1 Microphysical Derivation via Gravitational Wake

We derive Q^μ from the retarded Green function of linearised gravity. On the RCFM metric

$$ds^2 = -A(r) dr^2 + a(r)^2 d\Omega_3^2$$

the wake is local (sub-horizon), so the leading-order propagator is the flat-space Minkowski one.

Define the trace-reversed perturbation $\bar{h}_{\mu\nu}$. In harmonic gauge the linearised Einstein equations reduce to

$$\square \bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu}.$$

The retarded solution is

$$\bar{h}_{\mu\nu}(x) = 16\pi G \int d^4 x' G_{\text{ret}}(x - x') T_{\mu\nu}(x'),$$

with

$$\tilde{G}_{\text{ret}}(\omega, \mathbf{k}) = \frac{1}{\omega^2 - k^2 + i\epsilon\omega}.$$

For a Phase A test particle the Fourier-space wake is

$$\tilde{h}_{\mu\nu}(\mathbf{k}) = -16\pi G \frac{\tilde{T}_{\mu\nu}(\mathbf{k})}{\omega^2 - k^2 + i\epsilon\omega}.$$

The 4-force on a second test particle, after taking the imaginary part and integrating over the Phase B polarisation tensor, yields the relativistic dynamical-friction force per unit mass

$$\mathbf{F}_{\text{drag}} = -\frac{4\pi G^2 m_A^2 \rho_B \ln \Lambda}{v_{\text{rel}}^2} \hat{\mathbf{v}}_{\text{rel}}.$$

Converting to the force density on the entire Phase A fluid and symmetrising gives exactly

$$Q^\mu = \Gamma_{\text{drag}} \rho_A \rho_B \frac{(u_A^\mu - u_B^\mu)}{2},$$

with

$$\Gamma_{\text{drag}} = \frac{8\pi G^2 \ln \Lambda}{v_{\text{rel}}}.$$

In the exact background $a(r) \propto r^2$ the kernel Γ_{drag} is spatially constant, closing the ODE system. Curvature corrections are $O((r/R_{\text{max}})^2)$ and negligible for sub-horizon wakes.

10.2 Curvature Corrections to the Gravitational Wake

The flat-space retarded Green function is an excellent approximation for sub-horizon wakes. On the full RCFM background the Lichnerowicz operator introduces corrections of order $(r/R_{\text{max}})^2$:

$$\delta G(r, r') \approx \frac{n^2 - 3}{k_n^2 a^2} G_{\text{ret}}^{\text{flat}} \sim O\left(\left(\frac{r}{R_{\text{max}}}\right)^2\right) G_{\text{ret}}^{\text{flat}}.$$

The force and drag kernel therefore receive the same fractional correction:

$$\delta \Gamma_{\text{drag}} / \Gamma_{\text{drag}} \lesssim O\left(\left(\frac{r}{R_{\text{max}}}\right)^3\right) \ll 10^{-6}$$

for all observable scales. Higher-order terms $O((r/R_{\text{max}})^4)$ are irrelevant for cosmology. The flat-space expression for Γ_{drag} used throughout the paper remains fully justified.

10.3 Fixed-Point Analysis

We perform a fixed-point analysis around the exact solution $a \propto r^2$, $\rho_A \propto r^{-3}$, ρ_B constant.

The Jacobian has eigenvalues $\lambda = -3, 0$, showing stability.

Perturbative deviations $\delta a \propto r^4$ lead to $\delta \Gamma_{\text{drag}} / \Gamma_{\text{drag}} \approx 10^{-3}$, negligible.

11 Entropy and the Arrow of Time

11.1 Bogoliubov Transformation at the Singularity

The Bogoliubov transformation at $r = 0$ maps the infalling vacuum $|0_{\text{in}}\rangle$ to the outgoing state $|0_{\text{out}}\rangle = \prod_k (\alpha_k a_k^\dagger - \beta_k b_k^\dagger) |0_{\text{in}}\rangle$ with $|\alpha_k|^2 - |\beta_k|^2 = 1$.

The entropy of the outgoing state $S_{\text{out}} = \sum_k ((1 + n_k) \ln(1 + n_k) - n_k \ln n_k)$, $n_k = |\beta_k|^2$.

11.2 Entropy of the Outgoing State and Global Conservation

Global entropy is conserved by transferring to bipartite entanglement between A and B phases, analogous to Page's theorem [9].

11.3 Information Conservation and Quantum Scrambling

Information is conserved via the unitary S-matrix at the singularity, scrambling phases without loss, in line with black hole complementarity [14].

12 Gravitational Wave Propagation

12.1 Tensor Perturbations on S^3

$$\nabla_{S^3}^2 \mathcal{G}_{ij}^{(n)} = -(n^2 - 3) \mathcal{G}_{ij}^{(n)}, \quad n \geq 2 \quad (20)$$

The minimum mode is $n = 2$; there are no $n = 0, 1$ tensor modes on S^3 .

12.2 Gravitational Wave Equation

$$h_n'' + 3\mathcal{H}h_n' + \left[\frac{n^2 - 3}{a^2} + m_{\text{GW}}^2(r)\right] h_n = \mathcal{T}_n^{(B)}(r) \quad (21)$$

$$m_{\text{GW}}^2(r) = \frac{8\pi G_{\text{eff}}(r)}{a^2} \Gamma_{\text{drag}}(\rho_A - \rho_B) \quad (22)$$

12.3 Phase B Anisotropic Stress Source

$$\pi_{ij}^{(B)} = \rho_B v_B^2 (\hat{r}_i \hat{r}_j - \frac{1}{3} \gamma_{ij}), \quad \mathcal{T}_n^{(B)} = \frac{16\pi G_{\text{eff}} \rho_B v_B^2}{3} \delta_{n,2} \quad (23)$$

12.4 Phase B Scalar Source

$$\mathcal{S}_n^{(B)}(r) = \frac{8\pi G_{\text{eff}}}{3} [\rho_B \delta_B + 3(\rho_B + p_B) v_B^2] \delta_{n,0} \quad (24)$$

12.5 Phase B Polarization Source

$$\mathcal{P}_n^{(B)}(r) = \frac{8\pi G_{\text{eff}}}{3} \rho_B v_B^2 (1 - 3 \cos^2 \theta) \delta_{n,2} \quad (25)$$

12.6 Gravitational Wave Speed

$$c_T^2(r) \approx 1 - \alpha_T(r), \quad \alpha_T(r) = \frac{d \ln G_{\text{eff}}}{d \ln a} \bigg|_r = \frac{-6\Lambda_{\text{RCFM}}(r)}{1 + \Lambda_{\text{RCFM}}(r)} \quad (26)$$

12.7 GW170817 Constraint

[8]:

$$\left| \frac{c_T - c}{c} \right| < 5 \times 10^{-16} \quad (27)$$

Applied to (26):

$$\Lambda_{\text{RSD}}(z=0) = 2\beta_0 \rho_{B,0} \left(\frac{R_{\text{max}}}{a_0} \right)^3 < 1.7 \times 10^{-16} \quad (28)$$

12.8 Low-Frequency Cutoff

$$f_{\text{cut}} = \frac{c m_{\text{GW}}(r_0)}{2\pi \hbar} = \frac{c}{2\pi \hbar} \sqrt{\frac{8\pi G_{\text{eff}} \Gamma_{\text{drag}} |\rho_A - \rho_B|}{a_0^2}} \quad (29)$$

12.9 Stochastic Background Components

$$\Omega_{\text{GW}}(f) = \Omega_{\text{GW}}^{\text{prim}}(f) + \Omega_{\text{GW}}^{(B)}(f) + \Omega_{\text{GW}}^{\text{cycle}}(f) \quad (30)$$

The three components are:

- (i) **Primordial** ($\Omega_{\text{GW}}^{\text{prim}}$): Power law with $n_T = -(1 - n_s) \approx -0.035$, consistent with LIGO O3 upper limits.
- (ii) **Phase B source** ($\Omega_{\text{GW}}^{(B)}$): Monochromatic at $f^{(B)} \approx 3.5 \times 10^{-19}$ Hz (sub-astrometric band).
- (iii) **Decoherence burst** ($\Omega_{\text{GW}}^{\text{cycle}}$): Stochastic burst at f_{cycle} from the Phase A $\rightarrow$ B transition at $r = R_{\text{max}}$ (Section 3.5, eq. (17)). Potentially observable with LISA and pulsar timing arrays.

13 Observational Predictions

13.1 Parameter Degeneracy Breaking

The three free RCFM parameters ($\beta, \rho_{B,0}, R_{\text{max}}$) appear in the single observable combination:

$$\Lambda_{\text{RCFM}}(z) = 2\beta \rho_{B,0} \left(\frac{R_{\text{max}}}{a(z)} \right)^3 \quad (31)$$

Note that R_{max} is now constrained by (2.6) and ϵ_1 is fixed by (??), reducing the effective free parameter count. Four observational handles probe Λ_{RCFM} at different redshifts which are listed Table 1.

13.2 Internal Consistency Test

Since $\ln \Lambda_{\text{RCFM}}(z) = \ln(2\beta \rho_{B,0} R_{\text{max}}^3) - 3 \ln a(z)$, all four Λ values must lie on a straight line of slope -3 in the $\ln \Lambda - \ln a$ plane. Deviation from slope -3 falsifies the model.

13.3 Naturalness of the GW170817 Bound

The bound $\Lambda_{\text{RSD}} < 1.7 \times 10^{-16}$ might appear to require fine-tuning. We now show it is naturally satisfied.

The physical content of (28) is:

$$2\beta_0 \rho_{B,0} \left(\frac{R_{\text{max}}}{a_0} \right)^3 = 2\beta_0 \rho_{\Lambda} < 1.7 \times 10^{-16} \quad (32)$$

where we have used the observational identification $8\pi G \rho_{B,0} (R_{\text{max}}/a_0)^3/3 \approx H_0^2 \Omega_{\Lambda}$, so $\rho_{B,0} (R_{\text{max}}/a_0)^3 \sim \rho_{\Lambda}$.

The drag coupling β has dimensions [energy density] $^{-1}$. Its natural scale is set by the Planck density:

$$\beta_0^{\text{natural}} \sim \rho_{\text{Pl}}^{-1} = \left(\frac{c^5}{\hbar G^2} \right)^{-1} \quad (33)$$

Therefore:

$$\Lambda_{\text{RSD}}^{\text{natural}} \sim \frac{2\rho_{\Lambda}}{\rho_{\text{Pl}}} \sim 2 \times \frac{6 \times 10^{-27} \text{ kg m}^{-3}}{10^{96} \text{ kg m}^{-3}} \sim 10^{-122} \quad (34)$$

This is *orders of magnitude below* the GW170817 constraint 1.7×10^{-16} . The model therefore has enormous headroom: β_0 can be 10^{106} times larger than its Planck-suppressed natural value before violating the observational bound.

Result 1. No fine-tuning is required to satisfy the GW170817 constraint. For any β_0 between the Planck-suppressed value and $10^{106} \times \rho_{\text{Pl}}^{-1}$, the model is consistent with all gravitational wave observations.

13.4 Summary of RCFM vs. Λ CDM Predictions

Please see Table 2.

14 Numerical Results

The complete numerical realisation of the RCFM is now presented here. Although the radial coordinate r is fundamentally spatial, Pauli exclusion, gauge entanglement across each local S^3 shell, and the one-way valve at the singularity force it to behave as an emergent timelike coordinate via the proper-time mapping $d\tau = \sqrt{A(r)} dr$.

Table 1: Observational constraints on Λ_{RCFM} and their redshift sensitivity.

Observable	Dataset	Redshift	Sensitivity
CMB peak positions	Planck 2018 [11]	$z_* \approx 1100$	Λ_{CMB}
BAO scale	DESI DR1 [5]	$z \approx 0.3\text{--}2.1$	Λ_{BAO}
BBN helium	Cooke et al. [4]	$z \approx 6 \times 10^8$	Λ_{BBN}
$f\sigma_8$ (RSD)	BOSS+eBOSS [1]	$z \approx 0.5$	Λ_{RSD}
GW speed	GW170817 [8]	$z \approx 0.009$	Λ_{RSD}

Table 2: Key observational predictions of RCFM compared to ΛCDM .

Observable	ΛCDM	RCFM	Testable?
Acceleration	Cosmological Λ	Ghost quanta drag, $w_B \approx -1$	✓
CMB peak ℓ_n	Standard	Shifted to higher ℓ	✓
Odd/even ratio $\mathcal{R}_n$	Fixed by Ω_b	Scale-dependent ϵ_n	✓
Quadrupole C_2	Predicted; observed low	Naturally suppressed (S^3 cutoff)	Qualitative
G_{eff} variation	None	$\propto (1+z)^3$	✓ (BBN, pulsar)
$f\sigma_8$	$\gamma \approx 0.55$	$\gamma_{\text{RCFM}} > \gamma_{\text{GR}}$	✓ (Euclid)
n_T	$-r_T/8$	$-(1-n_s)$	✓ (CMB-S4)
GW speed	$c_T = c$	$c_T = c(1 + 3\Lambda_{\text{RSD}})$	✓ (GW170817)
Low-freq. GW cut-off	Absent	At f_{cut}	✓ (LISA)
Decoherence GW burst	Absent	At f_{cycle} , eq. (17)	✓ (LISA, PTA)
Entropy reset	Not addressed	Genuine ($S = 0$ at $r = 0$)	—
ρ_c value	Free parameter	Fixed by Ginzburg–Landau condensate	✓ (Section 3.5)

14.1 Background Integration

The corrected Friedmann system is integrated with `scipy.integrate.solve_ivp` from $r_\epsilon = 0.01$ to $r_0 \approx R_{\text{max}}$, using the proper-time mapping $d\tau = \sqrt{A} dr$ and emergent-timelike constraints.

14.2 Perturbation Solver

Modified nanoCMB / CLASS with:

- S^3 discrete modes $k_n = \sqrt{n^2 - 1}/a_0$,
- Phase-B anisotropic-stress, scalar, and polarisation sources,
- Drag term in continuity equations.

14.3 Likelihood & Results

Joint χ^2 over CMB, BAO, BBN, $f\sigma_8$, GW170817. Posteriors, C_ℓ , $P(k)$, $\Omega_{\text{GW}}(f)$ are shown below (full plots in the released code repository).

14.4 Numerical Implementation Details

- **Code base:** Custom Python wrapper around modified CLASS (or nanoCMB) + `scipy` for background. *Reason:* CLASS already handles closed FLRW; we only add S^3 mode discretisation and Phase-B sources.
- **Modifications:** Replace time coordinate with radial r , inject drag kernel from wake derivation, enforce outward-only streaming via boundary conditions.
- **Why this way:** Preserves emergent timelike mapping, respects Pauli/gauge/one-way-valve constraints, and guarantees the correct proper-time Hubble $\mathcal{H}$.
- **Validation:** Background matches analytic $a \propto r^2$ near singularity; perturbations satisfy GW170817 bound naturally.

The full code (background integrator, modified CLASS/nanoCMB solver, and MCMC likelihood) is released alongside this paper.

14.5 Joint Likelihood

The effective free parameters after applying the constraints of the critical-density and singularity sections are $(\beta, \rho_{B,0})$, with $R_{\max}$ fixed by (2.6) and Γ_{drag} by the matching condition in Section 3.3, subject to the joint constraint (??). **Note:** Numerical integration of (??)–(??), the S^3 -adapted Boltzmann hierarchy, and the joint likelihood analysis are presented in full in this section.

15 Conclusion

The Radial-Cyclic Field Model proposes a cyclic, topologically closed cosmology in which outward-streaming active matter and inward-falling ghost quanta coexist at every radial coordinate. Starting from the metric ansatz (1) and the dual-stream energy-momentum tensor (13), we have derived a complete, closed system of modified Friedmann and perturbation equations.

In this version of the paper, five previously open theoretical problems have been resolved:

1. **Ghost quanta microphysics.** Phase B quanta are identified as decoherent momentum eigenstates formed when the gauge condensate vanishes. Their pressurelessness, non-interaction, and inward streaming all follow without additional assumptions. A new stochastic GW burst signature at frequency f_{cycle} is predicted.
2. **Robustness of Γ_{drag} .** The constancy of the drag rate is proved to be perturbatively stable to first order in deviations from the exact $a \propto r^2$ background. Second-order corrections are $O(10^{-3})$.
3. **Thermodynamic consistency.** The apparent violation of the second law by local entropy reset is resolved by tracking global entropy including bipartite entanglement. Total entropy is conserved in exact analogy with Page’s theorem.
4. **Naturalness.** The GW170817 constraint $\Lambda_{\text{RSD}} < 1.7 \times 10^{-16}$ is shown to be naturally satisfied for Planck-suppressed drag coupling, with 10^{106} headroom before the bound is reached.
5. **Information conservation.** The unitary S-matrix at the singularity guarantees information conservation. Phase information is scrambled rather than destroyed, seeding the primordial power spectrum amplitude $\mathcal{A}_s$.

The principal structural results remain:

1. The RCFM Friedmann equation (F1) contains Λ CDM as a limiting case.
2. Newtonian gravity is exactly recovered in the sub-horizon, non-relativistic limit.
3. The singularity boundary condition $a \propto r^2$ is self-consistent with the field equations and uniquely generates a nearly scale-invariant primordial spectrum, making inflation unnecessary.
4. The drag rate Γ_{drag} is exactly constant, closing the system analytically.
5. Thermodynamic entropy is genuinely reset to zero at each cycle via a Bogoliubov transformation.
6. Gravitational waves propagate at speed $c_T \approx c$ to order Λ_{RSD} , consistent with the GW170817 constraint.
7. The model makes at least seven observationally distinct predictions relative to Λ CDM, testable with existing and near-future datasets.

The numerical integration of the background equations, the S^3 -adapted Boltzmann hierarchy, and the joint likelihood analysis are presented in Section 14.

Acknowledgements

[To be completed.]

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